Unit 1

1. Introduction to AI

Module 1.3

1.3 AI-Based and Conventional Systems

Tutorial 1.3

1.3 AI-Based and Conventional Systems

AI-1.3.1 K2 Differentiate between AI-based systems and conventional systems.

AI-Based and Conventional Systems

- In conventional computer systems, the software is programmed in programming languages created by humans, which makes it easy for humans to understand how the system transforms inputs into outputs
- In an AI-based system, machine learning (ML), patterns in data are used to determine how it should react in the future to the new data
- For example:
  - An AI-based image processor designed to identify images of cats is trained with a set of images known to contain cats
  - The AI determines on its own what patterns or features in the data can be used to identify cats
  - These patterns and rules are then applied to new images in order to determine if they contain cats

Difference between AI-Based and Conventional Systems

Feature	Conventional Systems	AI-Based Systems
Programming Style	Rule-based: explicitly coded instructions	Data-driven: learns patterns from data
Learning Ability	No learning; behavior is fixed	Learns and improves over time
Adaptability	Requires manual updates for changes	Adapts automatically to new data
Error Handling	Errors must be manually fixed	Can learn from mistakes and self-correct
Decision Making	Based on predefined logic	Based on probabilistic models and predictions
Data Dependency	Works best with structured data	Handles both structured and unstructured data
Transparency	Easy to understand and debug	Often a “black box” with less interpretability

THANK YOU!!!